DriveCompose: Observation-Anchored Joint Completion of Composable Gaussian Driving Worlds

Anonymous CVPR submission

Paper ID *****

Abstract

Driving logs observe only part of a scene, whereas simulation exposes surfaces through new camera positions and actor states. Completing these surfaces requires coordinating world-space background with canonical vehicles and articulated humans while retaining the recorded appearance. We present DriveCompose, which builds a composable Gaussian world by jointly completing typed canonical queries. A shared recovery front end supplies geometry, identity, and state evidence. Local query groups preserve each category's spatial structure, and global memory exchanges context across background and assets. Conditional flow fills unknown latent codes, restores observed codes after every step, and suppresses known free space. A fixed per-query decoder preserves the observed Gaussian parameters; a separate rendering check detects degradation caused by newly added content. Once accepted, the background and canonical assets remain fixed. Camera poses, rigid transforms, and skeletal poses then drive rendering across trajectories that share the same completion realization. This formulation places generation in offline world construction and makes subsequent scene changes explicit through state-driven composition.

1. Introduction

A driving log records the surfaces visible along one trajectory. Moving a simulated camera can expose a road patch hidden behind a vehicle; turning that vehicle reveals its unseen side; changing a pedestrian's pose exposes a different part of the body. A world built for such interventions must explain the recorded images and provide content for surfaces those images do not cover. The completed content must also remain attached to the same scene and actor identities as their states change.

Structured Gaussian representations provide a useful basis for this task. Street Gaussians separates tracked vehicles from the background, and OmniRe extends canonical modeling to articulated humans and other actor types [2, 10]. HUGSIM connects reconstructed scenes to control-driven state updates [15]. These representations distinguish persistent content from motion: a vehicle has one local shape, and a human has one canonical surface and skeleton binding. Completion must respect this distinction. Geometry generated for a particular view or pose must still have a well-defined place in the reusable world.

Generative methods address missing observations at several levels. StreetCrafter and HorizonForge combine three-dimensional conditions with video diffusion, while WorldSplat generates Gaussians and refines their rendered videos [8, 11, 16]. GenAssets learns a generative latent space for driving assets, and CompleteSplat completes Gaussian scenes from partial image evidence [5, 13]. A driving world brings these completion problems together. Its background is represented in world coordinates, vehicles in rigid local frames, and human surfaces through canonical templates. Independent completion leaves the road, neighboring actors, and asset placement to be reconciled after generation.

Our starting point is that a query needs two spatial descriptions. Its canonical coordinate identifies a persistent piece of content; its reference-world coordinate describes where that content lies relative to the surrounding scene. DriveCompose retains both descriptions and lets heterogeneous query groups exchange information during completion (Fig. 1). Reliable observations become fixed latent anchors, ray evidence identifies free space, and reserved supports represent unobserved content. Type-specific codecs put these supports into a common-width latent space. Local attention follows each support's structure, while global memory communicates scene context between groups.

Joint generation also couples the components through image formation. An unknown surface belongs to one asset, yet it can affect pixels already explained by another. A completed vehicle side can intersect a ray whose background is supported by the log. Observation preservation is therefore a scene-level property, even when the anchors and codec operate query by query. Locking an observed Gaussian's pa-

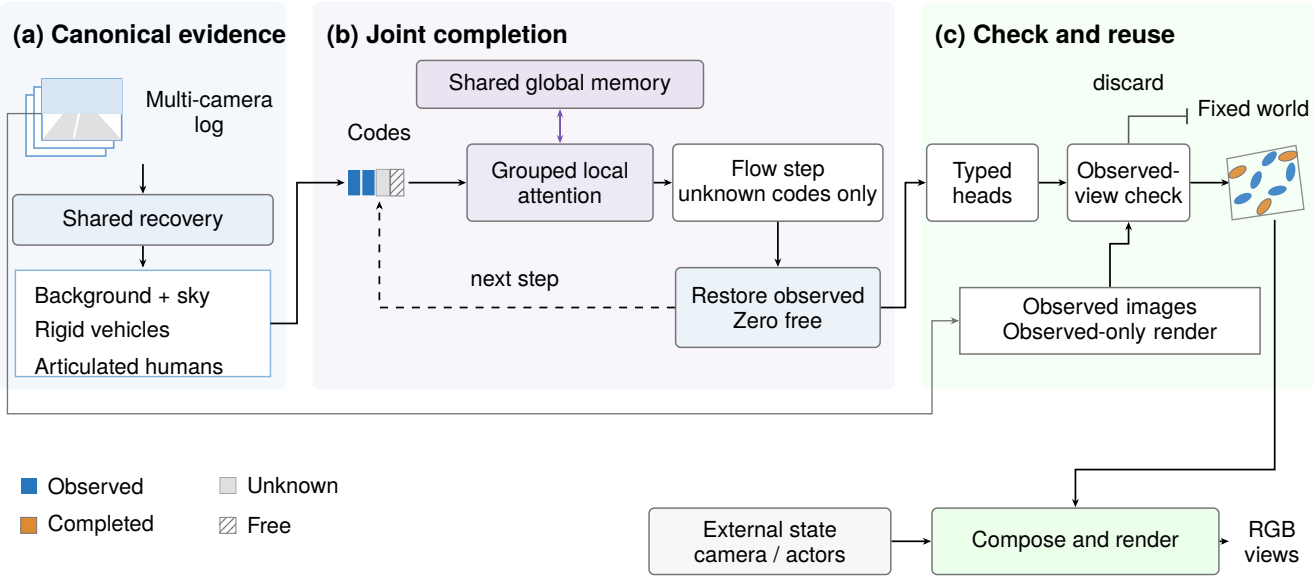


Figure 1. **DriveCompose construction and reuse.** (a) A shared front end recovers evidence on world-space and canonical supports. (b) Local query groups exchange context through global memory; each flow step restores observed codes and clears free space. (c) Typed heads decode the completed codes. An observed-view check compares the candidate with the images and an observed-only decode; rejected candidates do not enter the fixed world. External policy and dynamics supply the states used for subsequent composition and RGB rendering. All scene marks are schematic.

rameters leaves this occlusion route open; a penalty on those parameters cannot measure the effect of added content. We therefore handle parameter preservation and rendered consistency separately. Conditional flow updates only unknown codes, restores observed codes at every step, and clears free-space codes. A fixed per-query decoder transfers these constraints to Gaussian parameters. The decoded candidate is then rendered from the original cameras and compared with both the input images and an observed-only reference. This check measures local image degradation, foreground intrusion, absolute error, and coverage before accepting a completed world.

Acceptance separates construction from use. The world stores a fixed background and one canonical asset per identity. At runtime, the current state places the camera, applies rigid transforms to vehicles, and articulates humans through their fixed skeleton bindings. Geometry, covariance, and viewing directions are transformed consistently. External policy and dynamics update the state; resetting or branching a trajectory reuses the same world and completion seed. This separation makes the completed three-dimensional content, the current actor configuration, and the source of state changes explicit.

Our contributions are:

- A joint completion formulation for background and typed canonical assets, using canonical coordinates for persistent content and reference-world coordinates for scene context.
- An observation-anchored construction process that combines hard latent constraints and free-space suppression with candidate-versus-reference rendering checks.
- A fixed Gaussian world representation with consistent state-driven transforms for geometry, covariance, and directional appearance across reused trajectories.

2. Related work

Compositional reconstruction and simulation. Street Gaussians represents the background and tracked vehicles as separate Gaussian components [10]. OmniRe uses category-dependent canonical representations to include both rigid and articulated actors [2]. HUGSIM combines reconstructed content with interactive state updates and a simulation backend [15]. These works establish content/state separation and category-specific motion models. DriveCompose uses the same separation to define what completion should produce: a fixed background and canonical assets whose identities, local geometry, and binding persist across trajectories. The completion variables span these components jointly, rather than requiring their unobserved surfaces to be generated independently.

Feed-forward driving reconstruction. STORM, DGGT, StreetForward, and PointForward recover driving-scene representations from temporal observations [1, 3, 12, 14]. We use temporal recovery to establish the evidence that conditions completion: the front end predicts geometry, identity, and state, after which a fixed query layout records surface support, free space, and unknown regions. Conditional flow operates on the unknown codes during offline construction. Subsequent rendering reads the accepted content and current state, without repeating latent completion for each trajectory.

Generative scene and asset completion. StreetCrafter conditions video diffusion on LiDAR renderings and uses generated views to guide Gaussian reconstruction [11]; HorizonForge combines editable Gaussian–mesh content with video diffusion [8]. WorldSplat generates Gaussians followed by rendered-video refinement [16]. DriveCompose instead completes the content used directly by its state-driven Gaussian renderer. Among latent methods, GenAssets targets driving assets, CompleteSplat targets Gaussian scene completion, and JRM aggregates unaligned observations of a shared object through latent flow matching [5, 9, 13]. LatentWorld provides entity-centered control over semantic occupancy generation [7]. These methods use different generation units: an asset, a scene, or a shared object reconstructed from several observations. In DriveCompose, these units coexist in one query layout, retaining category-specific transforms and sharing an observed-view check. Our variables couple world-space background with rigid and articulated canonical assets. Observed/free constraints govern sampling and decoding, and the acceptance check controls how completed components enter one reusable RGB-rendering world.

3. Method

DriveCompose constructs a world once and renders it under changing states. Offline construction recovers evidence from a driving log, aggregates it on canonical supports, completes unknown latent codes, and checks the decoded content in the observed views. The accepted background and assets are then fixed; online composition uses only the current camera and actor states (Fig. 1).

3.1. Problem formulation and world representation

The input is a synchronized log of images $\mathbf{I}_{t,v}$, timestamps, camera intrinsics $\mathbf{K}_v$, and camera-to-rig calibration $\mathbf{T}^r_{c_v}$. Rig world poses $\mathbf{T}^w_r(t)$ are supplied by the log or estimated relative to its first frame. The camera world pose is $\mathbf{T}^w_{c_v}(t) = \mathbf{T}^w_r(t)\mathbf{T}^r_{c_v}$. Transforms act on column vectors, and distances are measured in meters. Images use linear RGB and a rectified pinhole model; distortion, rolling shutter, and exposure differences are handled before aggregation.

We construct a world

$$\mathcal{W} = \big(\mathcal{G}_b, C_{\text{sky}}, V_{\text{sky}}, \{\mathcal{A}_a\}, \mathcal{B}, \xi\big), \tag{1}$$

where $\mathcal{G}_b$ is the world-coordinate background, C_{sky} a cubemap with supported directions V_{sky}, $\mathcal{A}_a$ a canonical asset for identity a, $\mathcal{B}$ the finite spatial support, and ξ the completion seed. Each Gaussian stores a center $\boldsymbol{\mu}$, positive-definite covariance $\boldsymbol{\Sigma}$, opacity α, and first-order local directional appearance. Vehicles store fixed local geometry and dimensions. Humans additionally store a rest skeleton and skinning weights. These parameters remain fixed after acceptance.

Shared recovery. The Driving Log Encoder embeds camera calibration and time with image features. Cross-view/time summaries provide scene context. A trajectory-level slot identifies each actor, and state attention predicts its root pose and human joint rotations at each time. Geometry heads recover camera-axis depth, normals, confidence, semantics, and instance assignment. Actor poses are predicted in the rig frame and mapped to world coordinates with $\mathbf{T}^w_r(t)$. Each human has one shape code throughout the log, selecting a template with consistent canonical support and binding.

3.2. Canonical queries and evidence aggregation

A query contains its canonical position $\mathbf{q}_i$, cell scale $\boldsymbol{\delta}_i$, type k_i, local group g_i, and reference-world coordinate $\mathbf{q}^w_i$. Background supports combine reliable depth anchors with a finite world grid. Vehicle supports combine motion-normalized anchors with a local volume grid. Human queries lie on an identity-specific canonical surface; sky queries are directions grouped by cubemap face. The layout stays fixed during completion. The reference-world coordinate places a query at the first input time, including the human skinning transform, so local identity and scene placement enter the model separately.

To aggregate image evidence, we place the query at the actor state of each frame:

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \text{xyz}(\mathbf{T}^w_a(t)\bar{\mathbf{q}}_i), & k_i = \text{vehicle}, \\ \text{xyz}(\mathbf{T}^w_h(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{2}$$

Here $\bar{\mathbf{q}}_i$ is homogeneous, xyz extracts Euclidean coordinates, and $\mathbf{A}_i(t)$ is the weighted skinning transform. Sky projection uses camera rotation alone.

Projection gives camera-axis depth $z_{i,t,v}$ and an image location. Let $\boldsymbol{\phi}_{i,t,v}$, $d_{i,t,v}$, and $c_{i,t,v}$ be the sampled feature, predicted depth, and confidence. With tolerance $\eta = \eta_{\text{abs}} + \eta_{\text{rel}} d$, the normalized depth residual is $r = (d - z)/\eta$. A surface observation must be inside the image, have reliable depth and confidence, belong to the query's instance,

and satisfy $|r| \le 1$. Background queries also require static-background semantics. For validity $a_{i,t,v}$,

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r_{i,t,v}^2/2), \tag{3}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v} [\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{4}$$

The query confidence is the maximum among its accepted observations. A reliable ray votes for free space only if the whole support cell lies in front of the observed surface:

$$z_{i,t,v} + \tfrac{1}{2}\|\boldsymbol{\delta}_i\|_2 < d_{i,t,v} - \eta. \tag{5}$$

Sky features use reliable sky semantics, zero depth residual, and no free-space votes. Positive aggregation mass marks an observed query. Otherwise, enough free-space votes mark it free; the remainder is unknown. The mutually exclusive masks m_i, f_i, and u_i satisfy $m_i + f_i + u_i = 1$.

Type-specific encoders map $[\mathbf{h}_i; \mathbf{q}_i; \boldsymbol{\delta}_i]$ to observed codes $\mathbf{z}_i^{\text{obs}} \in \mathbb{R}^D$. Unobserved codes are zeroed before conditioning. Visibility flags and confidence enter the completion network separately, keeping the intrinsic codec definition shared between partial and complete targets.

3.3. Observation-anchored joint completion

Local groups and global memory. The Joint Completion DiT receives current and observed codes, masks, confidence, type, both coordinate descriptions, cell scales, and log summaries. Background groups follow spatial neighborhoods; each asset forms a group; sky groups follow cube-map faces. Local self-attention updates each group. Shared memory then reads all groups and returns scene context:

$$\mathbf{M}' = \mathbf{M} + \text{Attn}_r(\mathbf{M}, \widetilde{\mathbf{X}}, \widetilde{\mathbf{X}}), \tag{6}$$

$$\mathbf{U} = \widetilde{\mathbf{X}} + \text{Attn}_w(\text{LN}(\widetilde{\mathbf{X}}), \mathbf{M}', \mathbf{M}'). \tag{7}$$

$\widetilde{\mathbf{X}}$ concatenates locally updated query tokens; the three attention arguments are queries, keys, and values. A residual feed-forward update follows $\mathbf{U}$. Flow-time modulation enters local normalization, and memory is initialized from learned tokens and log summaries. This exchange lets groups use shared scene conditions while their queries retain separate coordinates and motion models.

Conditional flow. We learn a velocity field by flow matching [6]. A complete target $\mathbf{z}_1$ and Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ share the fixed query layout. For flow time $\tau \in [0, 1]$,

$$\mathbf{z}_\tau = m \odot \mathbf{z}^{\text{obs}} + u \odot [(1-\tau)\boldsymbol{\epsilon} + \tau \mathbf{z}_1]. \tag{8}$$

Free-space codes stay zero. Unknown queries have target velocity $\mathbf{z}_1 - \boldsymbol{\epsilon}$. With an optional loss selector s_i,

$$\mathcal{L}_{\text{flow}} = \frac{\sum_i u_i s_i \|\mathbf{v}_\theta(\mathbf{z}_\tau, \tau, C)_i - (\mathbf{z}_{1,i} - \boldsymbol{\epsilon}_i)\|_2^2}{D \max(1, \sum_i u_i s_i)}. \tag{9}$$

C collects observed evidence and query conditions. Every unknown query interpolated in Eq. 8 needs a valid complete target; s_i selects loss terms within that valid set.

Starting at $\tau = 0$, we take S Euler steps of size $\Delta\tau = 1/S$:

$$\mathbf{z}^{j+1} = \Pi_{m,f}\left[\mathbf{z}^j + \Delta\tau\, u \odot \mathbf{v}_\theta(\mathbf{z}^j, j/S, C)\right]. \tag{10}$$

$\Pi_{m,f}$ restores observed codes and clears free-space codes after every step. The seed selects one completion realization. Flow time indexes offline generation; simulation time indexes subsequent state changes.

3.4. Decoding and observation checks

Typed decoding. Background, vehicle, and human heads decode each query independently into K child Gaussians. A child center is

$$\boldsymbol{\mu}_{i,k} = \mathbf{q}_i + \tfrac{1}{2}\boldsymbol{\delta}_i \odot \tanh(\mathbf{b}_{i,k}). \tag{11}$$

Positive scales and normalized rotations determine covariance. Opacity is multiplied by $1 - f_i$; human children inherit their support query's skinning weights. A separate head predicts sky colors, retaining the observation-supported validity domain. Appearance stores a constant RGB term and three local directional coefficients per channel. With a fixed per-query codec, observed child parameters are unchanged by Eq. 10. Disabling non-observed queries gives the observed-only reference.

Rendered acceptance. New content can occlude that reference even when its parameters are unchanged. We compose candidate and reference at the original actor states, then render both from the input cameras. Let $e_{\text{cand}}(p)$ and $e_{\text{obs}}(p)$ be their mean absolute RGB errors at pixel p. We average the positive regression

$$\Delta e(p) = \max\{e_{\text{cand}}(p) - e_{\text{obs}}(p), 0\} \tag{12}$$

over reliable non-ego pixels, preventing an improvement elsewhere from cancelling local degradation. We also bound absolute RGB and valid-depth errors, view coverage, and foreground intrusion. Intrusion means a newly added surface lies in front of the predicted reference depth while being absent from the observed-only render; its rate is normalized over reliable non-sky geometry pixels. Acceptance combines aggregate and worst-view bounds with minimum coverage. Predicted depth supplies the reference for these consistency checks; independent geometric accuracy uses a separate reference.

3.5. Learning stages and objectives

Recovery and codec fitting. The first stage supervises geometry, semantics, instance assignment, identity, and state.

One trajectory-level matching assigns predicted slots to target identities across all frames. The second stage freezes recovery and trains the typed codec. Partial and more complete observations use the same query positions, types, order, and ownership. Rendering or set matching supervises the decoded representation; direct coordinate losses apply only when child correspondences are known.

Completion prior. The third stage freezes both recovery and codec, and trains the completion network with Eq. 9 on complete scene or asset targets. The fourth stage keeps those modules frozen and differentiates through flow integration, decoding, and rendering to adapt completion to partial real logs.

Observation-based adaptation. Write $\langle \ell \rangle_\Omega$ for a mean over valid elements in mask Ω. RGB and depth terms are masked L1 errors: $\mathcal{L}_{\text{rgb}} = \langle \|\widehat{\mathbf{I}} - \mathbf{I}\|_1/3 \rangle_{\Omega_I}$ and $\mathcal{L}_{\text{depth}} = \langle |\widehat{d} - d| \rangle_{\Omega_d}$. For accumulated opacity a_r on known-free rays, foot signed distance d_f (positive above ground), contact mask Ω_c, and penetration depth d_p (positive inside another actor),

$$\mathcal{L}_{\text{scene}} = \lambda_e \langle a_r^2 \rangle + \lambda_g \langle [-d_f]_+^2 \rangle + \lambda_c \langle |d_f| \rangle_{\Omega_c} + \lambda_p \langle [d_p]_+^2 \rangle, \tag{13}$$

$$\mathcal{L}_{\text{adapt}} = \lambda_r \mathcal{L}_{\text{rgb}} + \lambda_d \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{scene}}. \tag{14}$$

Here $[x]_+ = \max(x, 0)$; unmasked means run over supplied ray or geometry samples. Terms use only available observations and backend geometry. Free-ray opacity constrains added content along rays, beyond the per-cell free-space mask. Hard latent restoration and frozen decoding enforce the observed anchors. The adaptation objective measures the image and geometric effects of added content. The λ coefficients weight the active soft constraints. Encoder, codec, query-schema, and completion versions are recorded together so targets and sampling conditions use the same representation.

3.6. State-driven composition and reuse

At online step k, state s_k specifies the ego pose, active assets, vehicle transforms, and human root and joint poses. For a vehicle,

$$\boldsymbol{\mu}^w = \mathbf{R}\boldsymbol{\mu}^a + \mathbf{t}, \qquad \boldsymbol{\Sigma}^w = \mathbf{R}\boldsymbol{\Sigma}^a\mathbf{R}^\top. \tag{15}$$

For humans, the weighted skinning transform is $\mathbf{A}_i = \sum_j w_{ij}\mathbf{P}_j\mathbf{B}_j^{-1}$, where $\mathbf{B}_j$ and $\mathbf{P}_j$ are rest and current global joints in the root frame. The center becomes $\bar{\boldsymbol{\mu}}_i^w = \mathbf{T}_h^w\mathbf{A}_i\bar{\boldsymbol{\mu}}_i^h$. The combined local linear map $\mathbf{J}_i$ gives covariance $\mathbf{J}_i\boldsymbol{\Sigma}_i\mathbf{J}_i^\top$ and maps world viewing directions back with $\mathbf{J}_i^{-1}$ for local appearance evaluation. Nearly singular skinning transforms are rejected. The root pose controls overall placement; the root joint retains its rest transform.

The renderer composites the background, active assets, and valid sky. Policy actions and external dynamics advance the state:

$$o_k = \mathcal{R}(\text{Compose}(\mathcal{W}, s_k), \mathcal{C}_k), \tag{16}$$

$$s_{k+1} = F_{\text{dyn}}(s_k, \pi(o_k), \Delta t). \tag{17}$$

Resetting a trajectory changes state but preserves $\mathcal{W}$ and its seed. Finite spatial support and supported sky directions determine coverage. Road meshes and collision proxies share the visual assets' coordinates and are maintained by the physical backend.

4. Experiments

We study dynamic scene reconstruction from sparse temporal observations along two axes: recovering scene appearance at unobserved times and rendering the reconstructed scene from off-trajectory viewpoints. Temporal evaluation separates interpolation from short-horizon prediction, while spatial evaluation tests rendering quality as the camera moves away from the recorded trajectory.

4.1. Experimental Setup

Dataset and temporal sampling. We evaluate on a fixed subset of 40 sequences from the Waymo Open Dataset scene-flow validation split. We adopt the 20-frame temporal sampling pattern used in STORM and the evaluation resolution used in Instant NuRec [4, 12]. Each clip contains 20 consecutive frames at 10 Hz. We use the FRONT camera and provide four context images at zero-based frame indices $\{0, 5, 10, 15\}$. Temporal interpolation is evaluated at indices $\{1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14\}$, and temporal extrapolation at $\{16, 17, 18, 19\}$. The latter corresponds to prediction horizons of 0.1–0.4 seconds beyond the last observation. Context frames are excluded from temporal reconstruction scores.

Baselines and inference settings. The current comparison includes STORM, DGGT, and StreetForward using their released checkpoints without additional fine-tuning [1, 12, 14]. All methods observe the same camera and context timestamps and render the same target views. STORM uses calibrated cameras during reconstruction. DGGT and StreetForward estimate cameras and use context-based alignment and intrinsic-calibration adapters before rendering. DGGT is evaluated without diffusion refinement; its temporal adapter extends the last observed tracked motion for extrapolation. These settings define the implementations evaluated here.

Image and depth metrics. RGB predictions and reference images are evaluated at 240×160 pixels. We report PSNR and SSIM over the full image and within shared 2D

annotation-box masks. These masks define annotated regions, rather than motion-segmented foreground. For depth, we report AbsRel and δ_1 on valid projected LiDAR samples after per-frame least-squares scale alignment, following the metric formulation in Instant NuRec [4]. Frame scores are averaged within each window, window scores within each sequence, and sequence scores with equal weight across the 40 sequences.

Preprocessing and calibration. The current STORM pipeline resizes quarter-resolution JPEG images to 240×160; DGGT and StreetForward resize the original JPEG images to 518×350 and 504×336, respectively. Each method retains its input normalization. Camera alignment uses the four context observations. In temporal evaluation, one DGGT window uses context-frame LiDAR to resolve an otherwise indeterminate metric scale. In spatial evaluation, DGGT and StreetForward additionally use context-frame LiDAR scale calibration when camera motion is too small or trajectory alignment is unreliable. Calibration excludes the shared annotated regions and uses no target-frame depth or RGB.

4.2. Temporal Reconstruction and Prediction

Evaluation protocol. We traverse non-overlapping 20-frame windows throughout each selected sequence. Incomplete or temporally invalid windows are excluded using the shared data checks, yielding 357 valid windows for each method. We report interpolation, extrapolation, and all-target scores separately. The all-target result combines the 12 interpolated frames and four extrapolated frames in each window.

Quantitative comparison. Table 1 reports the measured baseline results. STORM and DGGT achieve all-target PSNR values of 26.191 and 26.056 dB, respectively, while StreetForward achieves 25.256 dB. All three methods obtain lower PSNR on extrapolation than on interpolation. Table 2 complements the full-image comparison with annotated-region scores and scale-aligned depth metrics.

4.3. Off-Trajectory View Synthesis

Evaluation protocol. We evaluate spatial extrapolation on the first 20-frame window of each of the same 40 sequences. Starting from the recorded camera trajectory, we apply fixed lateral offsets of ±1, ±2, and ±4 meters in the local camera frame while retaining camera orientation, intrinsics, and timestamps. Together with the original trajectory, this produces seven 20-frame videos per scene and 280 videos per method. This experiment follows the lateral-shift evaluation strategy of WorldSplat and the extrapolated-view evaluation motivation of PointForward [3, 16].

Table 1. Temporal reconstruction on 40 Waymo validation sequences using one FRONT camera. Each evaluated method covers the same 357 windows. PSNR is in dB; higher is better. † marks one window calibrated with context-frame LiDAR. MOSAIC has not yet been measured.

Method	Interpolation	Extrapolation	All targets
STORM	26.427	25.481	26.191
DGGT†	26.265	25.429	26.056
StreetForward	25.653	24.065	25.256
MOSAIC	–	–	–

Distributional evaluation. The shifted trajectories do not have paired reference images. We therefore compute FID and FVD against a fixed population of images and videos from the original recorded trajectories. FID uses the four context timestamps, yielding 160 images per offset and method, to assess spatial changes at observed times. FVD uses the consecutive frames 0–15 at 10 Hz, yielding 40 clips per offset and method and incorporating temporal interpolation. We use Inception image features for FID and I3D video features for FVD. Scores are computed over the pooled sample population for each offset; left and right shifts are reported separately. All methods share the same scenes and real reference population. Lower scores indicate smaller feature-distribution differences; paired visualizations assess scene-specific structure and viewpoint response.

Quantitative comparison. Table 3 reports both metrics. STORM obtains the lowest FID and FVD at the one- and two-meter offsets, whereas StreetForward obtains the lowest values at both four-meter offsets. The ordering therefore depends on the displacement. The original-trajectory scores provide a reference for changes under viewpoint shifts; the spatial results use their own calibrated rendering pipeline and sample scope, rather than the all-window temporal evaluation.

Table 2. All-target image and depth metrics on the same 357 windows. Region scores use shared 2D annotation boxes, not motion-segmentation masks. Depth is evaluated after per-frame scale alignment. The DGGT calibration exception is the same as in Table 1.

Method	Full image		Annotated regions		Sparse depth	
	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	AbsRel ↓	δ_1 ↑
STORM	26.191	0.833	22.083	0.654	0.129	0.867
DGGT†	26.056	0.809	20.809	0.603	0.168	0.779
StreetForward	25.256	0.814	21.844	0.636	0.122	0.860
MOSAIC	–	–	–	–	–	–

Table 3. Off-trajectory synthesis on 40 FRONT-camera clips. FID uses 160 images per trajectory; FVD uses 40 videos of 16 consecutive frames. Each score compares against the same original-trajectory reference population. Left and right denote lateral camera displacement. MOSAIC scores are unmeasured.

FID ↓

Position	STORM	DGGT	SF	Ours
Original	44.56	38.45	50.01	–
Left 1 m	62.50	81.59	92.80	–
Right 1 m	61.44	66.39	75.79	–
Left 2 m	82.92	98.76	111.70	–
Right 2 m	84.52	87.96	94.92	–
Left 4 m	127.56	129.80	125.64	–
Right 4 m	137.58	126.22	121.52	–

FVD ↓

Position	STORM	DGGT	SF	Ours
Original	503.59	707.81	665.42	–
Left 1 m	717.19	939.82	856.76	–
Right 1 m	727.35	1028.18	901.14	–
Left 2 m	1014.69	1189.00	1073.96	–
Right 2 m	1044.03	1205.05	1116.08	–
Left 4 m	1688.40	1710.13	1560.61	–
Right 4 m	1719.92	1613.26	1596.65	–

5. Discussion

DriveCompose covers a finite region under fixed illumination. Its observation anchors inherit errors in recovered depth, identity, and canonical pose. Vehicles remain rigid; human geometry uses fixed clothing volume and skeleton binding, so poses outside the template's support can expose missing content. Reference-world coordinates describe the construction scene, while contact and collision validity after an asset moves are handled by the physical backend. The observed-view check constrains consistency with recorded evidence; geometry beyond that evidence and behavior under new states require their own assessment.

6. Conclusion

DriveCompose constructs a reusable Gaussian driving world by jointly completing background and canonical asset codes. Canonical supports preserve each category's geometry and motion model, while reference-world coordinates let the completion network use shared scene context. Hard latent constraints retain observed parameters and suppress free space; an observed-view rendering check evaluates the effects of added content before acceptance. The accepted world then stays fixed as camera placement, vehicle motion, and human articulation change through explicit state transforms. This construction/use separation gives multiple trajectories a common completed scene, with observation consistency checked at construction and state changes handled during composition.

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